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PORTFOLIO & ENGINEERING SHOWCASE · 2025–2026

Solutions I've designed, built and shipped

BI Scientist • AI/ML Engineer • Python Developer

Precious Ikebude — BI Scientist building AI/ML systems · github.com/ikebude

ABOUT ME

Economics foundation. BI precision. Production AI/ML.

Junior Data Analyst at 3Consulting Limited (Lagos) building commercial compliance, fraud, and intelligence systems alongside open-source AI/ML products.

AI / ML & MLOps

- XGBoost, LightGBM, Scikit-learn, Isolation Forest
- Imbalanced learning (PR-AUC, recall@k%, isotonic calibration)
- SHAP explainability, WoE credit scorecards, BG/NBD CLV
- MLflow Registry, Apache Airflow DAGs, PSI drift checks

BI, Data & Analytics

- Power BI (DAX, Power Query, star-schema modelling)
- SQL (PostgreSQL, MS SQL Server, DuckDB, SQLite)
- Python (Pandas, NumPy, SciPy, Statsmodels, Plotly)
- Cohort retention, RFM segmentation, executive KPI reporting

Software, APIs & Infra

- FastAPI, Flask, NestJS (TypeScript), REST & Webhooks
- React, Next.js, Electron desktop apps, Tailwind CSS
- Docker Compose, Kafka, Redis, MinIO, GitHub Actions CI
- Linux VPS, Oracle Cloud Free ARM, InMotion cPanel/SSH

PORTFOLIO MAP

Everything I've built — Company vs. Personal

Company platforms are shown strictly as engineering case studies (no proprietary source code). Personal products are open-source and reproducible.

Industry Solutions (3Consulting Ltd.)

- SafeData (safedata.3consult-ng.com) — NDPA/GDPR data retention & consent-withdrawal disposal platform
- FraudGuard (fraudguard.3consult-ng.com) — Real-time payment & login fraud scoring + analyst SAR queue
- Africa Stablecoin Intelligence (africastablecoinintelligence.com) — Pan-African stablecoin market & regulatory tracker

Personal Products & Open-Source Hub

- Moniguard v2 — Nigerian PoD & e-commerce fraud engine (PR-AUC 0.877, FastAPI 47.9ms p95)
- Lolo's Ledger — Multi-asset trading journal, risk calculator & AI coach (521/521 unit tests passing)
- LoanLens · Customer Intelligence · BI Operations Agent · AIOps Expense Tracker
- Hub Repo (ikebude/ai-ml-data-engineering) — 11 ML, SQL & Power BI projects with full Git history

01

Industry solutions at 3Consulting

Commercial compliance, fraud prevention, and market intelligence platforms built in Lagos. Shown as engineering case studies; source code stays private.

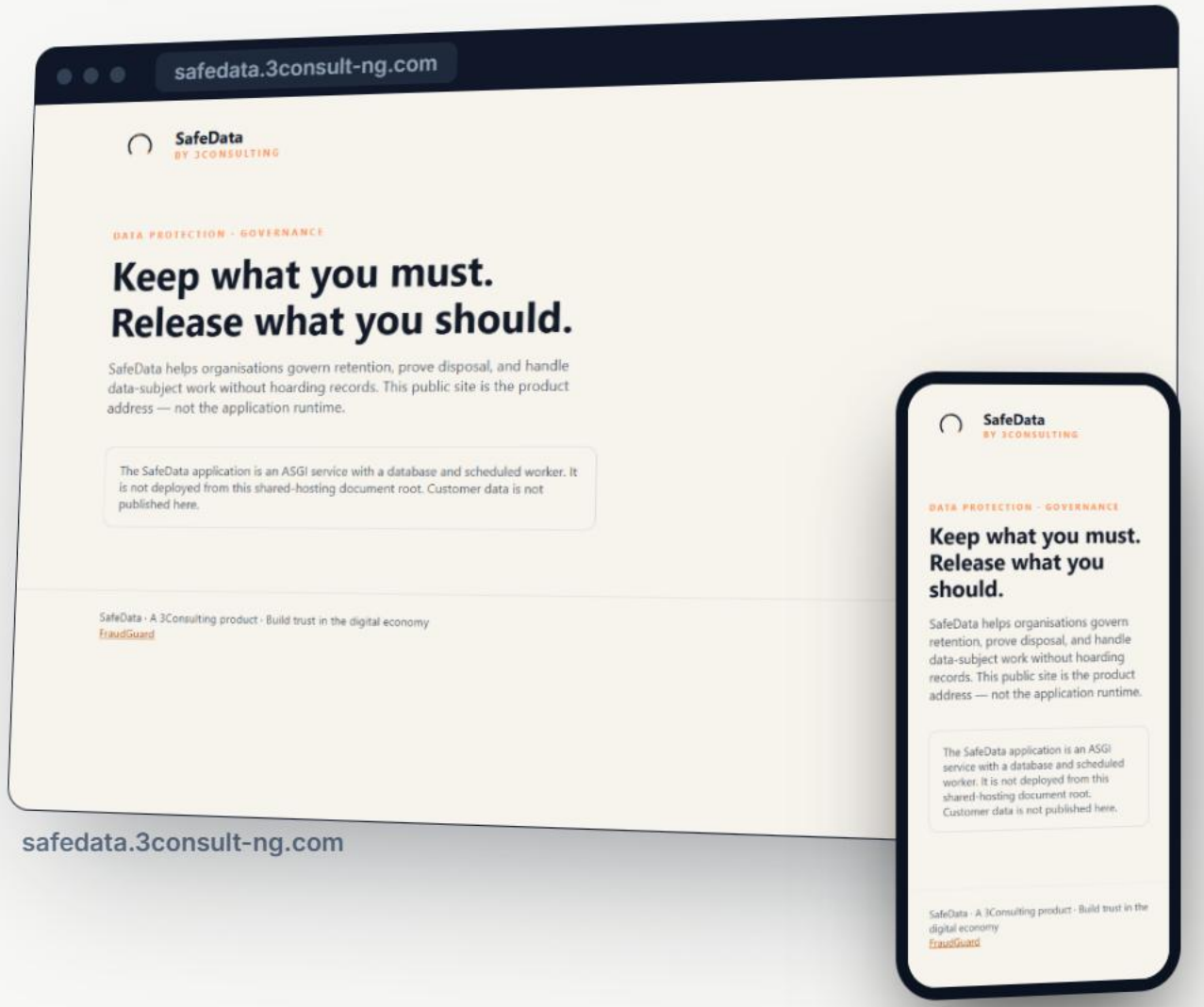
CASE STUDY 1 · DATA GOVERNANCE

SafeData

Keep what you must. Release what you should. Automated data retention and consent-withdrawal disposal.

- Tracks data categories past active use through to verifiable disposal
- Connects via API & webhooks to trigger disposal countdowns on consent withdrawal
- Enforces legal holds and dual approvals before high-risk records are purged

FastAPI | PostgreSQL | NDPA / GDPR



safedata.3consult-ng.com

Two unified engines: Retention & Consent

Designed so organisations govern data lifecycle compliance without hoarding sensitive personal records.

1. Retention Tracking

- Maps records to jurisdiction-aware retention schedules (NDPA, GDPR, sector rules)
- Calculates exact disposal eligibility dates once business purpose concludes
- Supports dry-run simulations before executing irreversible purges

2. Consent Withdrawal

- Receives HMAC-signed consent withdrawal webhooks from client CRMs/portals
- Immediately transitions affected subject records into a tracked disposal window
- Overrides disposal automatically if an active litigation/legal hold applies

3. Governance Registers

- Built-in RoPA (Record of Processing Activities), DSAR, DPIA & Breach logs
- Role-Based Access Control (Admin, DPO, Data Owner) with TOTP 2FA
- Tamper-evident SHA-256 hash-chained disposal certificates

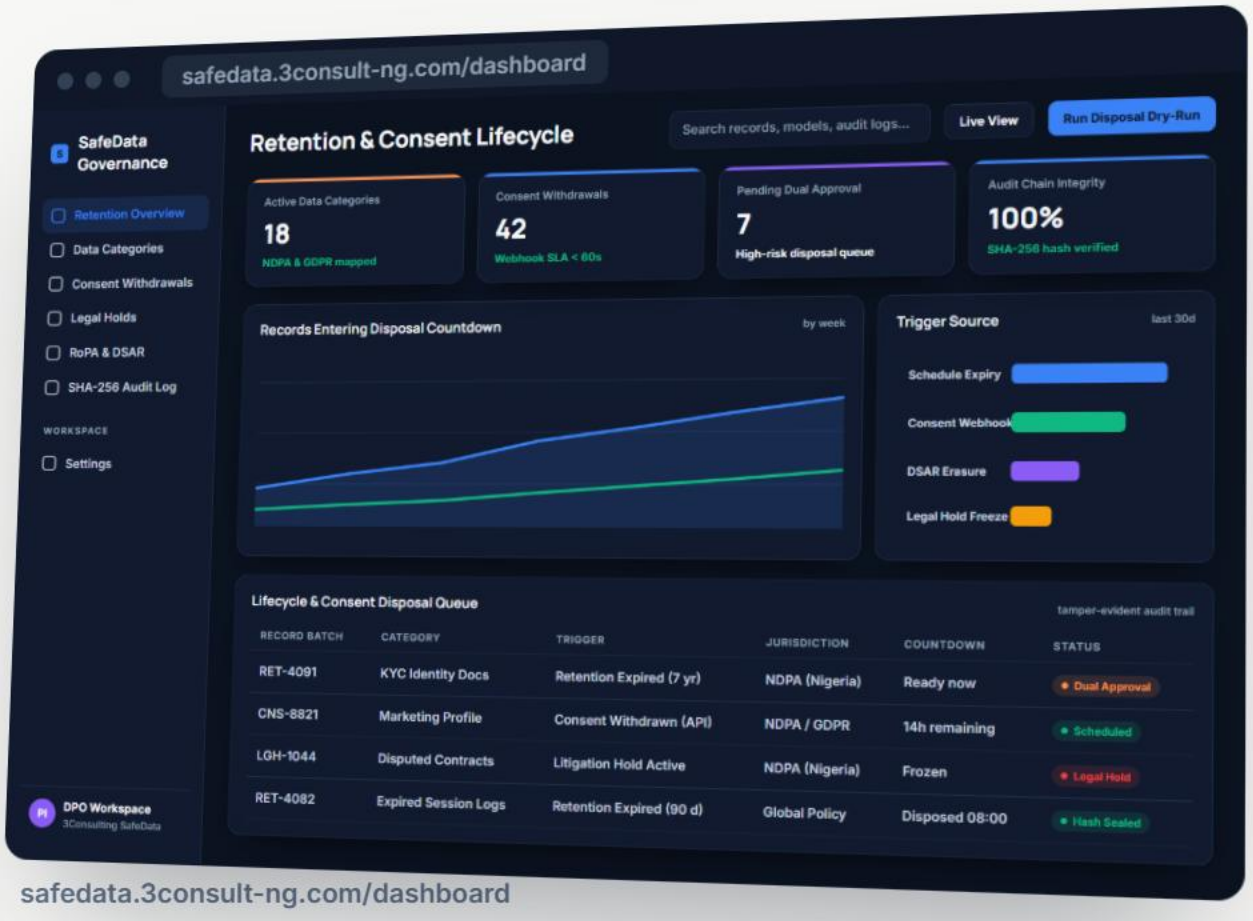
SAFEDATA · DPO WORKSPACE

Auditable disposal queues & legal holds

Every retention expiry and consent withdrawal is tracked from intake to hash-sealed disposal.

- Live countdowns for records awaiting single or dual DPO approval
- One-click litigation hold freeze across affected categories
- Multi-currency commercial onboarding with localised pricing

SQLAlchemy | Alembic | SHA-256 Audit



safedata.3consult-ng.com/dashboard

SAFEDATA · MY ENGINEERING CONTRIBUTIONS

Database reliability, CI security & lifecycle logic

Focused on making the FastAPI + PostgreSQL backend verifiable, tenant-safe, and deployment-ready.

Core Modules	Governance Registers	Audit Integrity	CI Pipeline
2	4	SHA-256	4 Gates
Retention + Consent Suite	RoPA · DSAR · DPIA · Breach	Chained cryptographic log	Lint · PG Tests · Deps · Secrets

Backend & PostgreSQL Test Suite

- Resolved PostgreSQL foreign-key & tenant-scoping test fixtures across the test suite
- Verified background worker job scheduling for automated retention scans

CI/CD & Deployment Runbooks

- Authored GitHub Actions workflow running PostgreSQL service containers, linting & secret scans
- Prepared cPanel subdomain + Linux ASGI VPS deployment runbooks for safedata.3consult-ng.com

SAFEDATA · LIVE REFERENCE

Lawful retention and provable disposal — built for NDPA and GDPR compliance.

Public showcase: https://safedata.3consult-ng.com · Case study: github.com/ikebude/ai-ml-data-engineering/tree/main/industry-work/safedata

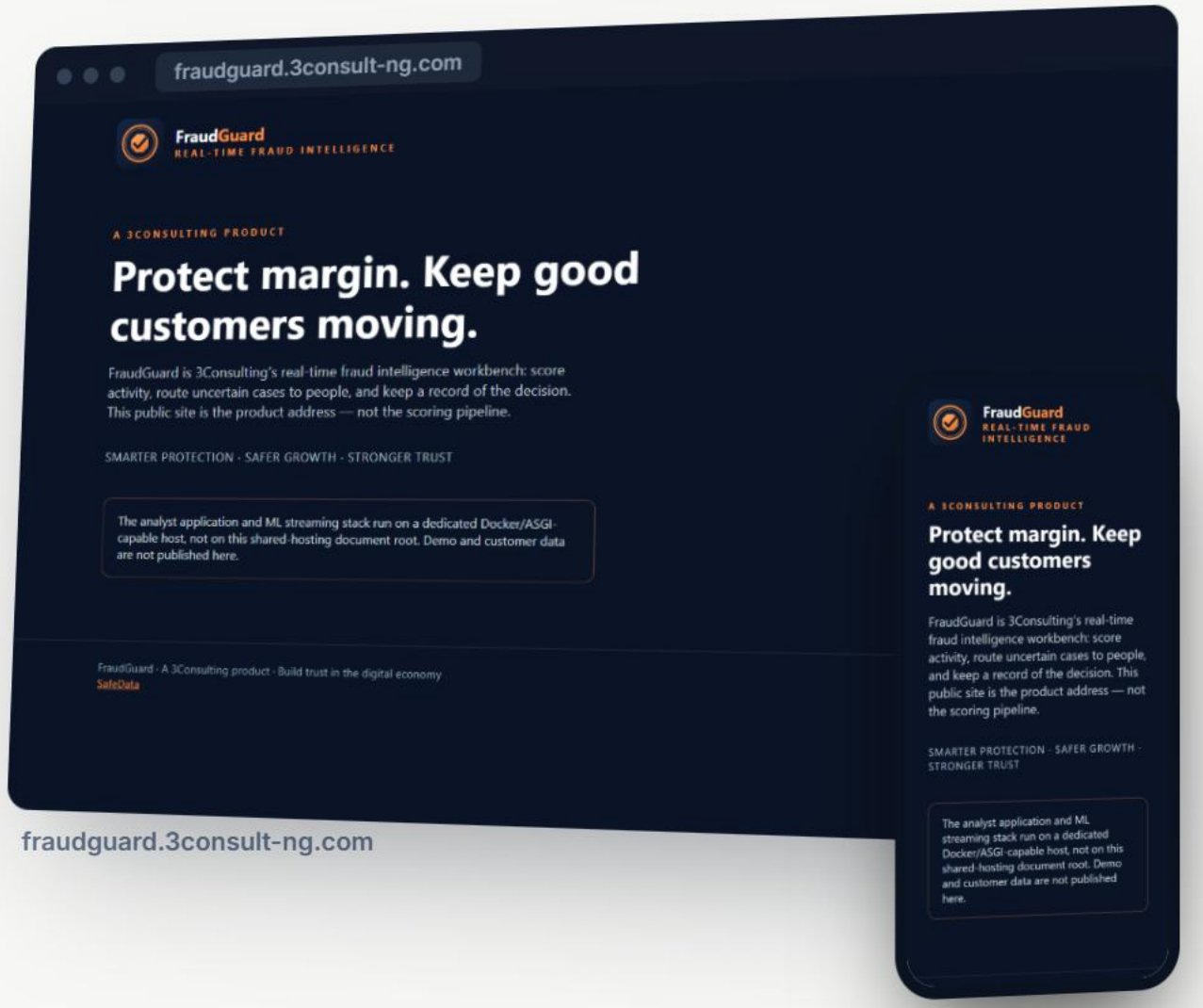
CASE STUDY 2 · FRAUD INTELLIGENCE

FraudGuard

Real-time transaction scoring, analyst triage queues, and continuous MLOps retraining.

- Scores payments, transfers, and logins 0–1000 in real time
- Auto-approves low risk, blocks high risk, and routes uncertain cases to analysts
- Captures analyst decisions to retrain models and generate SAR compliance packs

Python · Kafka · MLflow & Airflow



fraudguard.3consult-ng.com

FRAUDGUARD · ANALYST OPERATIONS

Prioritised triage with plain-English risk reasons

Analysts see why a transaction was flagged, linked devices/IPs, and SAR escalation controls.

- Flask operations workspace with role-based queues and audit trails
- Top feature drivers surfaced per transaction for fast human review
- Built-in Suspicious Activity Report (SAR) triage and GDPR workflows

Flask | PostgreSQL | XGBoost + IsoForest



fraudguard.3consult-ng.com/transactions

FRAUDGUARD · SYSTEM TOPOLOGY

Event-driven streaming & MLOps architecture

Designed around a transactional outbox and single canonical prediction writer to prevent data loss or training-serving skew.

1. Outbox & Kafka Backbone

- Producer writes raw events to PostgreSQL + transactional outbox table
- Outbox relay publishes reliably to Apache Kafka topics
- Prevents dual-write inconsistencies when downstream services restart

2. Canonical Streaming Inference

- Streaming consumer computes velocity & shared-entity features
- Loads champion XGBoost + Isolation Forest model from MLflow/MinIO
- Writes canonical fraud predictions & reason codes back to PostgreSQL

3. Airflow & Observability

- Airflow DAGs run scheduled retraining, bias checks, and retention jobs
- Prometheus, Grafana, and Alertmanager track latency and PSI drift
- Gated on-demand inference API serves synchronous explanations

From single-stack deployment to multi-industry packs

Commercial-readiness architecture and pilot validation roadmap authored for 3Consulting.

Shadow-Mode Pilot Protocol

- Ingest live client events in shadow mode before enabling active blocking
- Measure precision, recall@k% alert budget, and false-positive cost on client traffic
- Rotate all local defaults and enforce strict secret scanning prior to exposure

Pluggable Industry Packs

- Domain feature packs tailored for E-Commerce PoD, Mobile Money, and Core Banking
- Tenant-scoped API keys, webhook callbacks, and configurable decision thresholds
- Public reference: https://fraudguard.3consult-ng.com

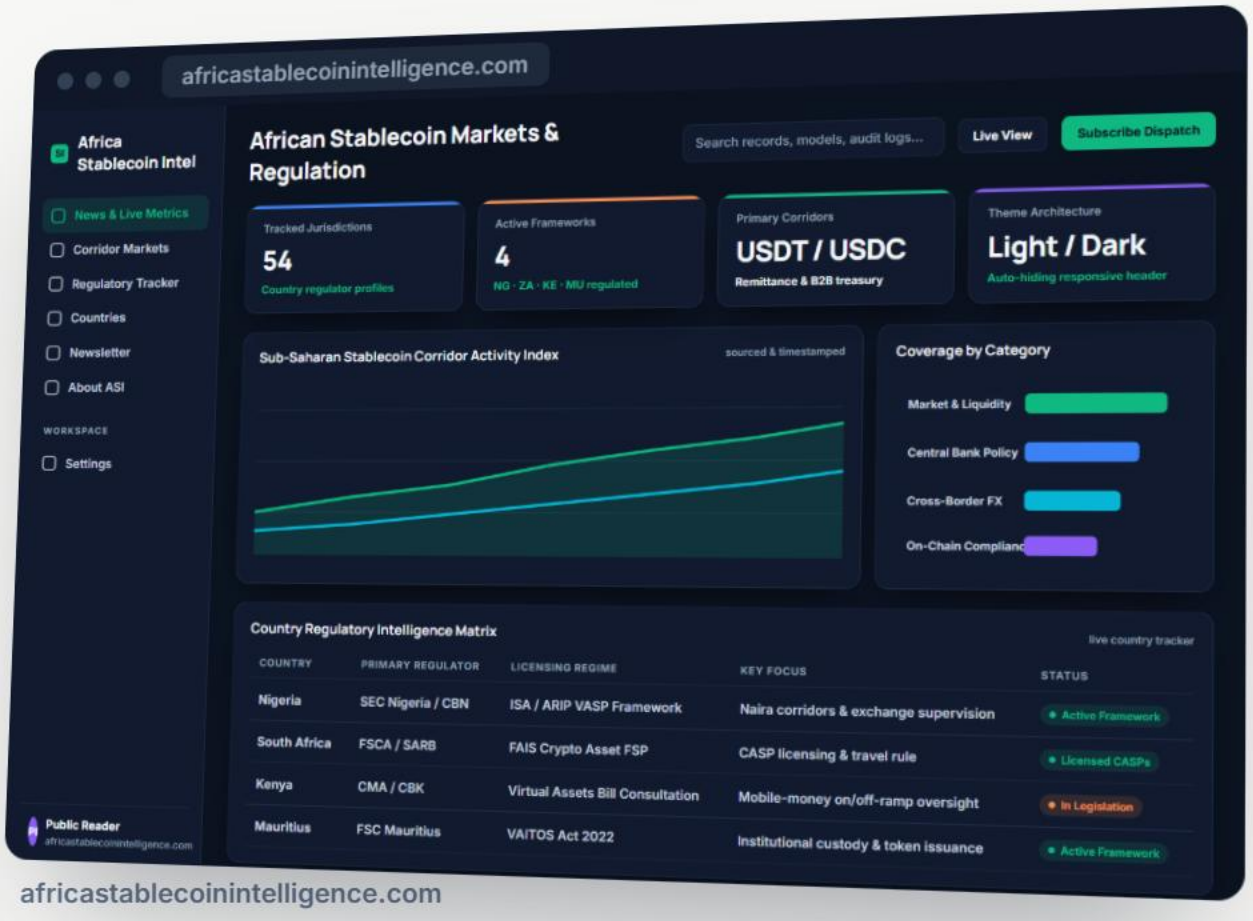
CASE STUDY 3 · MARKET INTELLIGENCE

Africa Stablecoin Intelligence

Tracking stablecoin adoption, corridor liquidity, and central-bank regulation across Africa.

- Rolling live market indicators with explicit source & timestamp provenance
- Dedicated Market News and Regulatory Watchdog dispatches
- Country-by-country regulatory intelligence directory across 54 jurisdictions

React / Next.js | Market Data | Light / Dark UI



africastablecoinintelligence.com

AFRICA STABLECOIN INTELLIGENCE · COVERAGE

Structured intelligence across markets & regulators

Separating high-frequency market/corridor activity from high-impact regulatory policy shifts.

Corridor & Market Metrics

- Tracks USDT/USDC adoption across remittance and B2B trade corridors
- Visualises country usage trends, on/off-ramp liquidity, and fraud alerts
- Moving Highlights card in the hero surfaces key weekly shifts

Country Regulatory Tracker

- Profiles central bank and SEC posture across African nations
- Highlights the 4 countries with active stablecoin/VASP frameworks today
- Tracks bills, circulars, and licensing updates as they impact operators

Editorial CMS & Launch Ops

- Role-gated admin panel for drafting, editing, and publishing dispatches
- Light/Dark/System theme switch persisted cleanly across public & admin apps
- Google Search Console verification and production deployment runbooks

AFRICA STABLECOIN INTELLIGENCE · LIVE PLATFORM

Bringing clarity to African digital-asset corridors and regulation.

Live at https://africastablecoinintelligence.com · Case study: github.com/ikebude/ai-ml-data-engineering/tree/main/industry-work/africa-stablecoin-intelligence

02

Personal AI & ML products

Independently designed, trained, and shipped open-source systems — built with real evaluation metrics and reproducible pipelines.

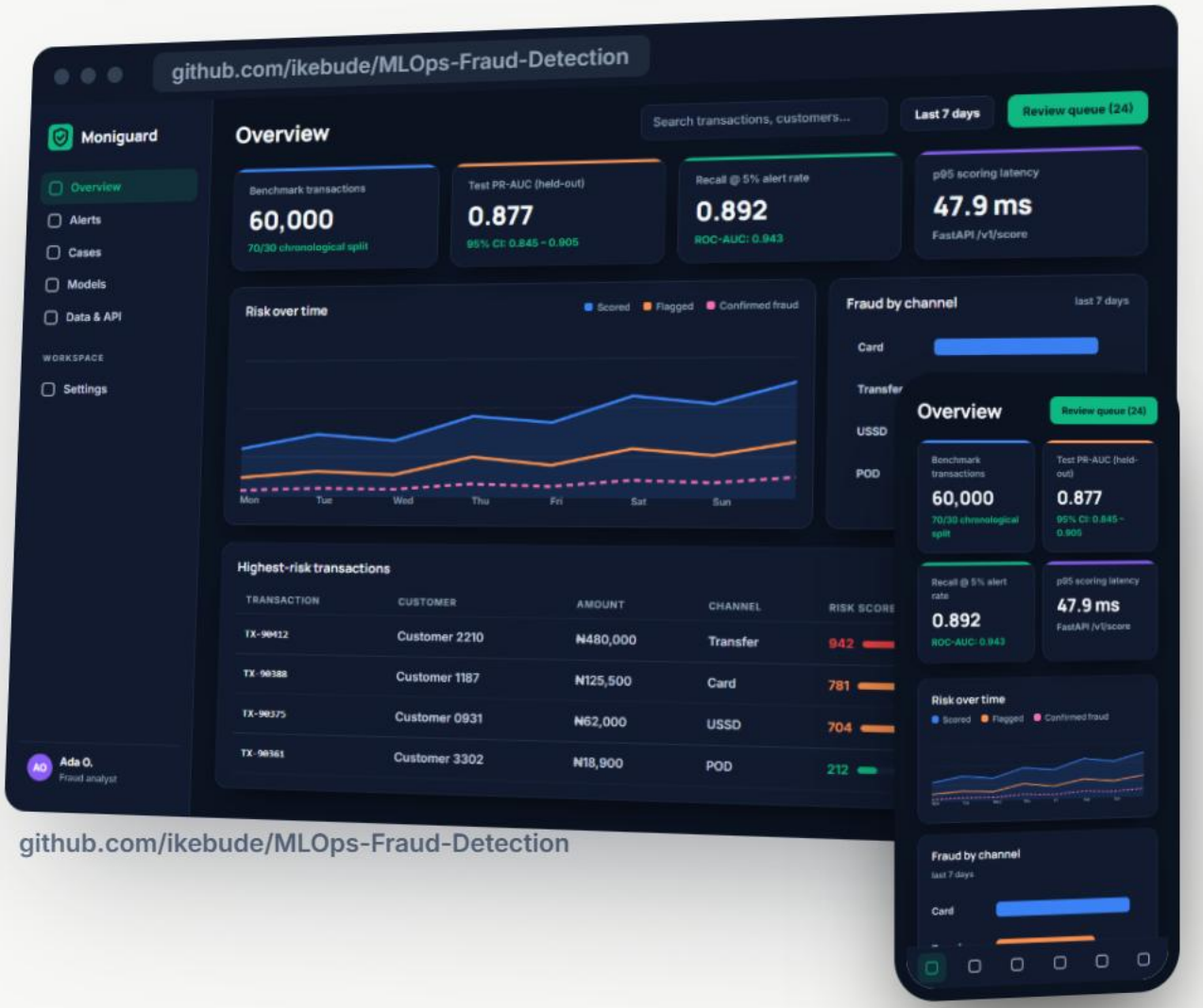
PERSONAL FLAGSHIP · FRAUD MLOPS

Moniguard v2

End-to-end fraud scoring for Nigerian pay-on-delivery e-commerce and mobile money.

- Single shared FeatureEngineer (74 velocity, behavioural & graph features) across train and serve
- Calibrated 70/30 XGBoost + Isolation Forest ensemble outputting 0–1000 scores
- FastAPI POST /v1/score endpoint returning score, decision & top reason codes

XGBoost | FastAPI | Airflow + MLflow



github.com/ikebude/MLOps-Fraud-Detection

MONIGUARD V2 · REPRODUCIBLE BENCHMARK (REPORTS/METRICS.JSON)

Measured on a strict 70/30 chronological split

60,000 synthetic Nigerian PoD transactions across 180 simulated days (42k train, 18k held-out future test; ~2.66% fraud across 6 patterns). Reproduce with make demo.

Test PR-AUC	Test ROC-AUC	Recall @ 5% Alerts	p95 Scoring Latency
0.877	**0.943**	**0.892**	**47.9 ms**
95% CI: 0.845 – 0.905	95% CI: 0.927 – 0.958	Recall @ 1% alerts: 0.377	Single-txn FastAPI /v1/score

Why Chronological Split + PR-AUC?

Random splits leak active fraud rings across train/test; time-based splitting tests future generalisation

At 2.6% fraud, raw accuracy is a vanity metric — PR-AUC and recall at fixed review capacity measure real ops value

Next Milestones (Documented Honestly)

Benchmark pipeline on public PaySim (6.3M mobile-money txns) and IEEE-CIS datasets via Kaggle API

Swap feature-importance reason weights for exact per-transaction SHAP TreeExplainer values

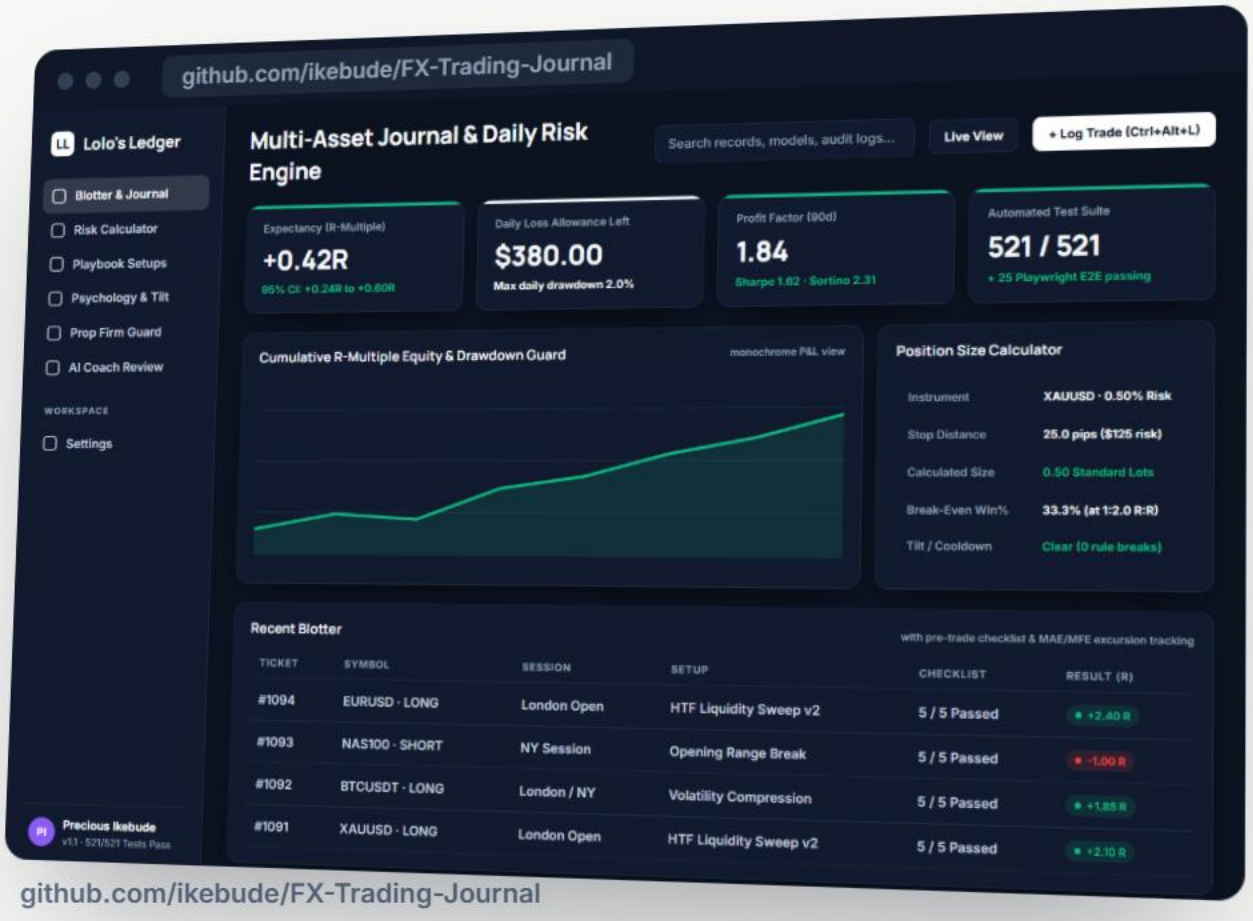
PERSONAL PRODUCT · FINTECH & RISK DISCIPLINE

Lolo's Ledger

Multi-asset trading journal, risk calculator, and educational AI coach (Desktop v1.1 + Web Monorepo).

- 521/521 unit tests + 25 Playwright E2E tests passing in shipped FXLedger v1.1 desktop core
- Tracks R-multiples, expectancy with 95% CIs, Sharpe/Sortino, MAE/MFE, and session heatmaps
- Multi-broker CSV/HTML importers, MT4/5 & Binance connectors, and tamper-evident audit hash chain

TypeScript | React + NestJS | 521 Unit Tests



github.com/ikebude/FX-Trading-Journal

LOLO'S LEDGER · RISK ENGINE & AI GUARDRAILS

Enforcing math and psychology before every trade

Built on pnpm + Turborepo (@ll/core, NestJS API, BullMQ worker, PostgreSQL/Redis, React web + Electron desktop).

Position & Daily Loss Calculator

- Calculates exact lots/units from account balance, risk %, and stop distance across FX, crypto, indices & stocks
- Displays trades left today and money left to lose before daily drawdown circuit breaker triggers
- Shows break-even win rate vs actual historical setup win rate

Behavioural & Tilt Detection

- Pre-trade checklist gate, mood/anxiety tracking, and post-loss advisory cool-down timers
- Automated detectors for revenge trading, size escalation, and overtrading
- Auto-captures multi-timeframe OHLC trade snapshots via headless lightweight-charts

Strict AI Coach Guardrails

- Never generates buy/sell signals or price predictions — strictly reviews the user's own journal data
- Offline Whisper voice-memo transcription & screenshot OCR degrade gracefully without network calls
- Every AI insight carries 'Educational, not financial advice' verification

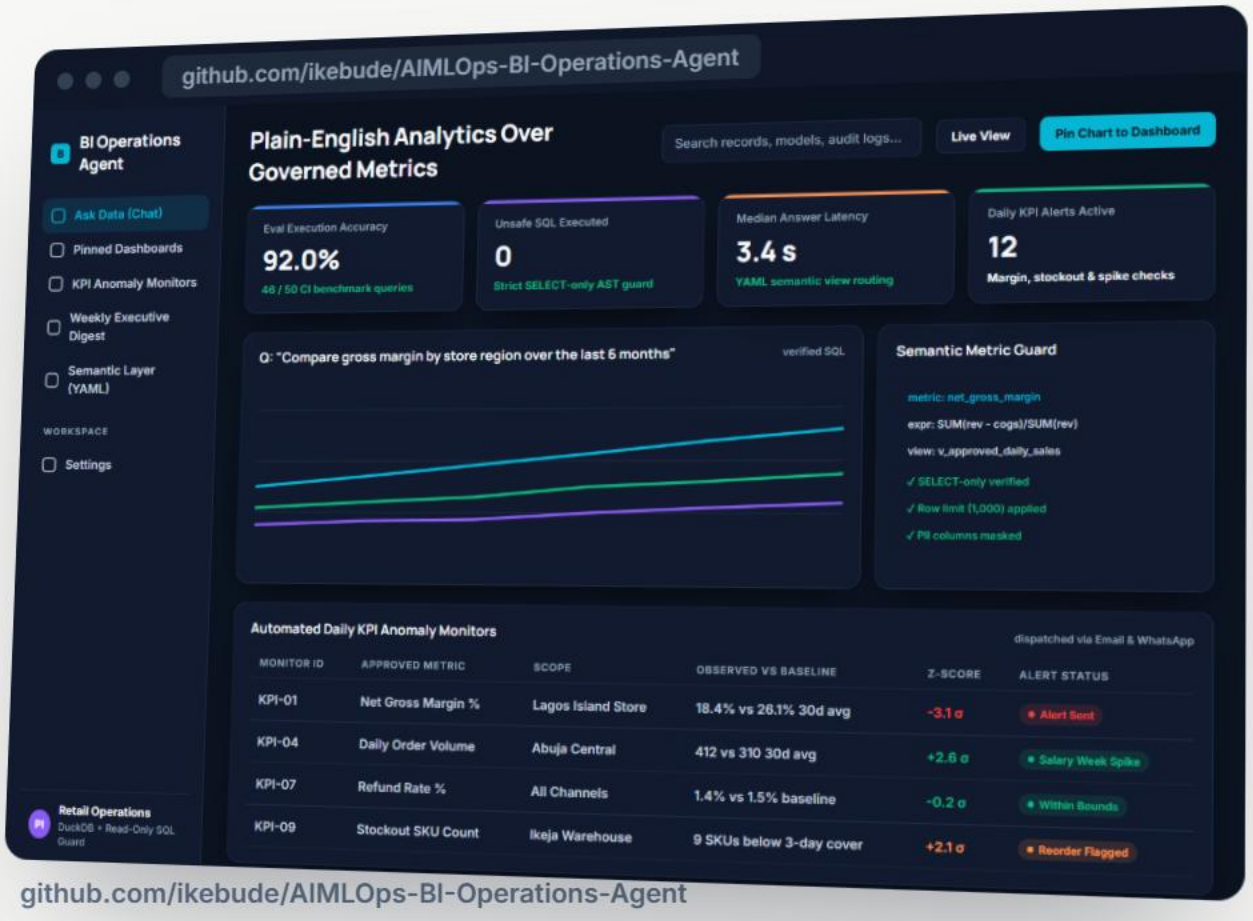
PERSONAL PRODUCT · LLM ANALYTICS & SEMANTIC GUARDRAILS

BI Operations Agent

Ask business data plain-English questions over a governed YAML semantic layer.

- LLM writes SQL strictly against approved metric views — never raw unvalidated tables
- AST SQL guard enforces SELECT-only, table allow-lists, row limits, timeouts, and PII masking
- Daily KPI anomaly monitors and weekly executive summaries evaluated on a 50-pair CI benchmark

FastAPI | DuckDB & Postgres | LLM Tool Loop



github.com/ikebude/AIMLOps-BI-Operations-Agent

Precious Ikebude · BI Scientist building AI/ML systems

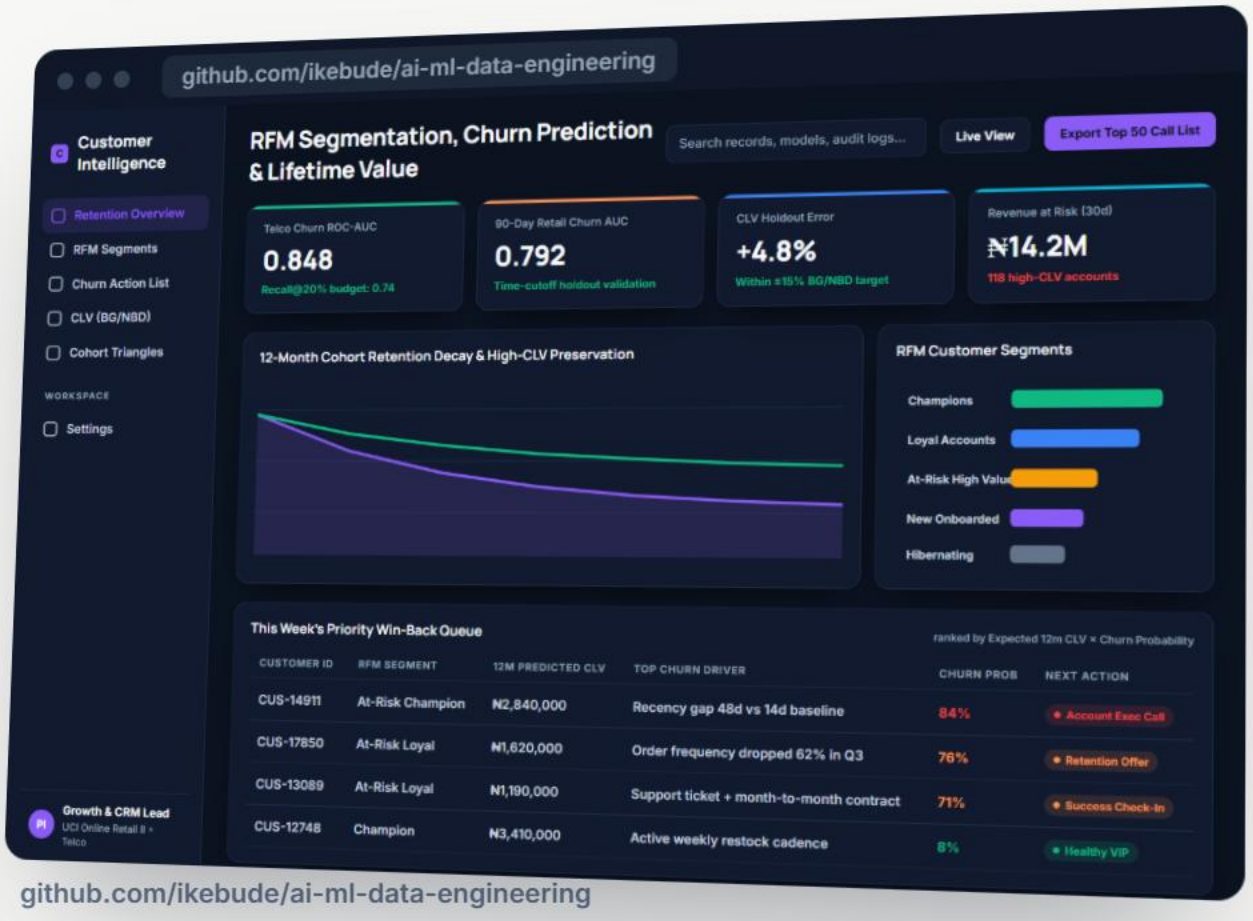
PERSONAL PRODUCTS · CREDIT, RETENTION & FINANCE ML

Credit, Retention & Finance ML

Turning exploratory notebooks into interactive decision products.

- LoanLens: Prosper P2P default scorecard (WoE + LightGBM) & approval simulator
- Customer Intelligence: RFM segments, churn action list & BG/NBD CLV backtest
- AIOps Expense Tracker: NGN finance app with ML categoriser & receipt OCR

LoanLens | Customer Intel | Expense Tracker

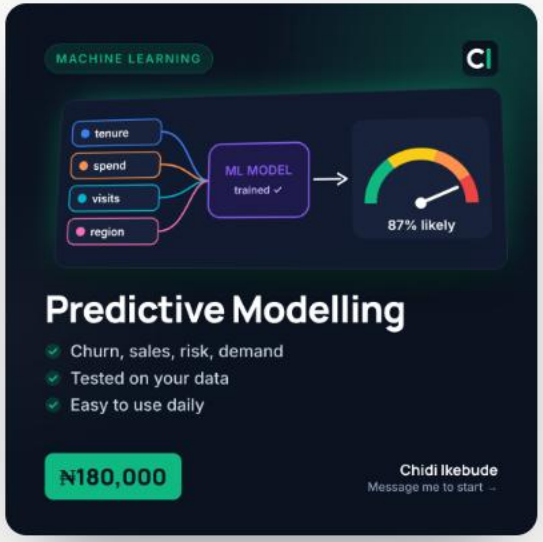


github.com/ikebude/ai-ml-data-engineering

11 ML, SQL & Power BI projects + client services



End-to-End BI & Power BI Dashboards



Churn, Credit Risk & Fare ML Pipelines



S&P 500, Prosper Loan & CRM EDA



MS SQL Server & PostgreSQL Engineering

ENGINEERING STANDARD

How I build, evaluate and ship

Every project in my portfolio follows strict data-honesty, testing, and security rules.

1. Honest ML Evaluation

- Time-based or group-aware splits — never random-shuffling temporal events
- PR-AUC, recall at fixed alert/contact budgets, and 95% bootstrap confidence intervals
- Calibrated probabilities + SHAP / scorecard reasons + documented model cards

2. Reproducible MLOps & CI

- Shared feature engineering modules imported by both training and live inference
- GitHub Actions CI running pytest unit/integration suites and lint checks on every push
- Docker Compose stacks that seed demo data, train, and serve with make demo

3. Security & Governance

- Zero hardcoded credentials — environment templates (.env.example) and secret scanning
- Read-only SQL guards for LLM agents, hashed API keys, and tamper-evident SHA-256 logs
- Strict separation between proprietary employer systems and open-source personal code

LET'S CONNECT

Precious Ikebude — BI Scientist building AI/ML systems

Available for AI/ML Engineer, BI Scientist, Data Scientist, Data Engineer, and Data Analyst roles (Lagos · Hybrid · Remote).

Direct Contact & Links

GitHub Hub: github.com/ikebude/ai-ml-data-engineering

GitHub Profile: github.com/ikebude

Portfolio: datascienceportfol.io/ikebude

Email: precious.ikebude@outlook.com · preshebere99@gmail.com · +234 813 678 7181

What I Bring to Your Team

Production ML & MLOps: from raw transaction/customer tables to Dockerised FastAPI/Airflow systems

Decision-First BI: executive Power BI dashboards, automated SQL pipelines, and clear business storytelling

Credentials: B.Sc. Economics (UNN) · Google Advanced Data Analytics · Microsoft Azure AI Engineer (In View)